Software Test Plan (STP)

Student Record System

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Contents

1. Introduction **2**
- Purpose . . . 2
- Scope . . . 2
- References . . . 2
- Definitions . . . 3

2. Test Items **3**

3. Features to be Tested **3**

4. Features Not to be Tested **4**

5. Test Approach / Strategy **4**
- Levels . . . 4
- Types . . . 6
- Entry Criteria . . . 6
- Exit Criteria . . . 6
- 5.1 Security Validation . . . 7

6. Test Environment **7**
- Hardware . . . 7
- Software . . . 7
- Tools . . . 7
- Test Data . . . 7

7. Test Schedule **8**
- Milestones . . . 9

8. Test Deliverables **9**

9. Roles and Responsibilities **9**

10. Risks and Mitigation **9**

11. Assumptions & Dependencies **10**

12. Suspension & Resumption Criteria **10**
Suspend testing if: 10
Resume testing if: 10

13. Test Case Management & Traceability **10**

14. Test Metrics & Reporting **12**
Metrics collected 12
Reports 12

15. Approvals **12**

1. Introduction

Purpose

This document defines the Software Test Plan (STP) for the Student Record System v1.0. It outlines test objectives, scope, strategy, target resources, execution schedule, and responsibilities required to verify system quality and functional compliance prior to production release.

Scope

Testing covers core software modules of the Student Record System:

- **Account Provisioning & Profile Services:** Administrative user provisioning, temporary credential display, mandatory password reset, and contact info self-service updates.
- **Course Setup, Self-Enrollment & Schedules:** Course registration, section capacity limits, rubric scheme weight auto-normalization, student self-enrollment (capped at 6 active courses), and class schedule publishing.
- **Digitized Attendance Marking:** Instructor binary roll-call entry, aggregate percentage tracking, 48-hour automated attendance edit freeze, and administrative overrides.
- **Weighted Rubric Grading Workflow:** Rubric score evaluation, grade state transitions (`Draft` → `Submitted` → `Under Review` → `Published`), and regrade dispute workflows.
- **Analytics & Summary Reporting:** Consolidated student GPA calculation, class grade distributions, and executive administrative metrics export.
- **System Non-Functional & Security Attributes:** API latency (95% $<$ 800 ms), 200 concurrent transactions, 99.0% operational availability, TLS 1.3 transit encryption, bcrypt password hashing, 48-hour JWT expiration, and RBAC endpoint middleware.

Hardware maintenance and third-party email/SMTP delivery infrastructure are excluded.

References

- **SRS:** Software Requirements Specification for Student Record System v1.0.
- **SAD:** Software Architecture and Design Specification for Student Record System v1.0.
- **IEEE Std 829-2008:** IEEE Standard for Software and System Test Documentation.

- **OWASP Top 10:** Web Application Security Verification Standard.

Definitions

- **API:** Application Programming Interface.
- **GPA:** Grade Point Average.
- **JWT:** JSON Web Token (HTTP-only bearer session authorization).
- **RBAC:** Role-Based Access Control.
- **RTM:** Requirements Traceability Matrix.
- **SRS:** Software Requirements Specification.
- **STP:** Software Test Plan.

2. Test Items

- **User Authentication & Account Module (`AuthModule`):** User provisioning, temporary credential validation, mandatory password resets, and profile management.
- **Course Registration & Schedule Module (`CourseModule`):** Section creation, faculty mapping, rubric component weight auto-normalization, and timetable scheduling.
- **Student Enrollment Module (`EnrollmentModule`):** Self-enrollment workflow enforcing the hard 6-course limit per student per term.
- **Attendance Engine (`AttendanceModule`):** Digitized roll-call entry, percentage aggregation, 48-hour automated edit freeze, and administrative overrides.
- **Grade Lifecycle & Dispute Engine (`GradingModule`):** Rubric score evaluation, state transitions (`Draft` → `Submitted` → `Under Review` → `Published`), and regrade dispute tracking.
- **Analytics & Summary Reporting Engine (`AnalyticsModule`):** Student GPA aggregation, class score distribution rendering, and administrative CSV metric export.
- **Security & Authorization Middleware (`SecurityMiddleware`):** TLS 1.3 transit encryption, bcrypt password hashing, 48-hour JWT expiration, and endpoint RBAC.

3. Features to be Tested

Features mapped directly to SRS requirement IDs:

- `SYS-F-001:` Admin provisioning of student, faculty, and admin user accounts with temporary credentials.
- `SYS-F-002:` Mandatory password reset upon initial login using temporary credentials.
- `SYS-F-003:` Student and faculty profile contact details updates.
- `SYS-F-004:` Admin updates to user account status and role metadata.
- `SYS-F-005:` Course section creation with automatic rubric weight normalization to 100%.
- `SYS-F-006:` Student course self-enrollment capped at a maximum of 6 active courses.
- `SYS-F-007:` Class schedule creation and timetable publishing.
- `SYS-F-008:` Faculty course assignment and roster data isolation.
- `SYS-F-009:` Digitized binary roll-call attendance submission by faculty.
- `SYS-F-010:` Automated 48-hour modification freeze on attendance sessions post-start.
- `SYS-F-011:` Administrative attendance record override and audit logging.
- `SYS-F-012:` Student view of aggregate attendance percentages per course.
- `SYS-F-013:` Faculty rubric-based mark entry in `Draft` state.
- `SYS-F-014:` Faculty grade submission applying weighted rubric normalization.

- **SYS-F-015:** Administrative publication of final student course grades.
- **SYS-F-016:** Student regrade dispute filing transitioning grade state to `Under Review` (permitted only once per grade; a repeat request is rejected).
- **SYS-F-017:** Faculty dispute resolution and grade re-publication (the re-published grade is final).
- **SYS-F-018:** Consolidated student GPA calculation and dashboard rendering.
- **SYS-F-019:** Faculty class grade distribution analytics summary.
- **SYS-F-020:** Administrative export of institution-level performance metrics to CSV.
- **SYS-NF-001:** Response time $\leq$ 800 ms for 95% of API requests under 500 active sessions.
- **SYS-NF-002:** Support for 200 concurrent attendance/grade submission transactions without database deadlock.
- **SYS-NF-003:** 99.0% system availability during active academic terms.
- **SYS-NF-004:** Responsive web interface (375px to 1920px) with $\leq$ 3 click roll-call navigation.
- **SYS-NF-005:** Modular 3-tier architecture maintaining $\geq$ 70% automated test coverage.

4. Features Not to be Tested

- Central University Single Sign-On (SSO) / OAuth identity provider internals (assumed tested & managed by university IT)
- University database server infrastructure and hardware host maintenance (cloud provider / IT responsibility)
- Third-party SMTP mail server agent delivery logic (email transport vendor responsibility)

5. Test Approach / Strategy

Levels

Testing executes across four hierarchical verification levels:

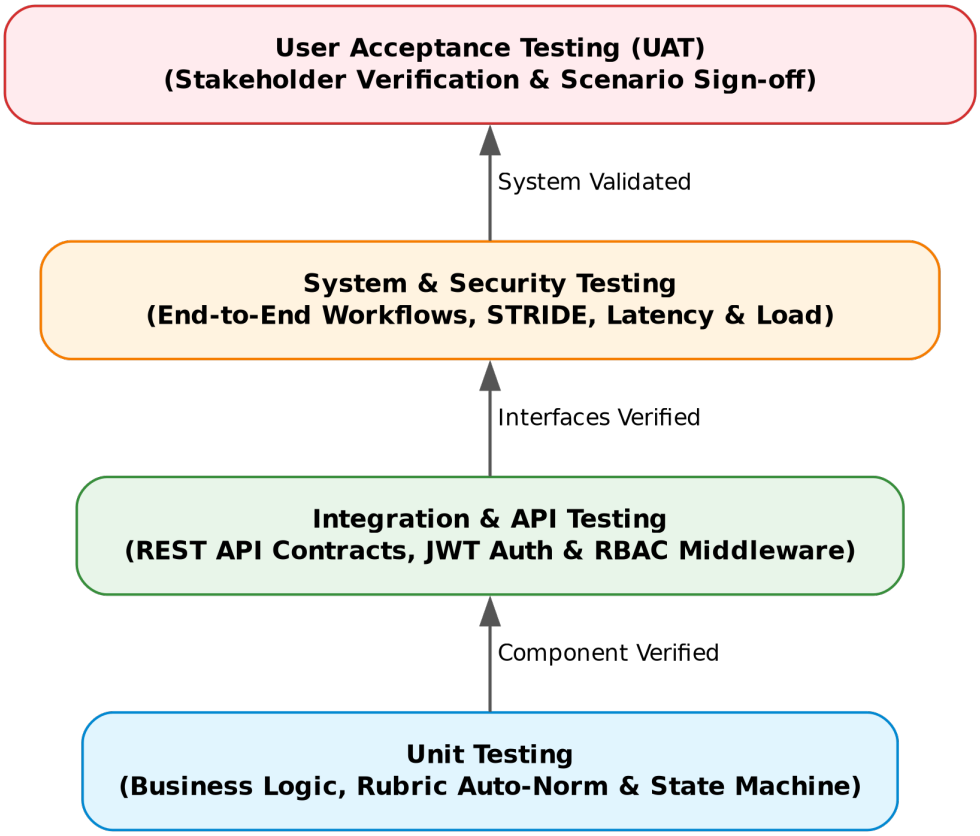


Figure 1: Testing Levels Hierarchy

- **Unit Tests:** Verify component functions, rubric weight auto-normalization algorithms, GPA calculation logic, and state machine transition rules.
- **Integration Tests:** Verify REST API endpoints, ORM data mappers, authorization middleware, and relational database queries.
- **System Tests:** Validate full end-to-end user workflows across Presentation, Application, and Persistence layers.
- **Acceptance Tests (UAT):** Conduct scenario verification with evaluators for Student, Faculty, and Admin roles.

Types

- **Functional Testing:** Core business logic verification against functional requirements `SYS-F-001` through `SYS-F-020`.
- **Regression Testing:** Automated test execution on every pull request and on every push to `main` (GitHub Actions) to ensure existing functionality remains unbroken.
- **Performance Testing:** Load testing verifying API response time ≤ 800 ms under 500 active sessions and 200 concurrent transactions.
- **Usability Testing:** Accessibility and workflow efficiency evaluation (≤ 3 clicks to complete roll-call).

Entry Criteria

- Baseline application build deployed to the test environment.
- Database schema migrated and seeded with synthetic test data.
- Test suites and API scripts verified against OpenAPI specifications.

Exit Criteria

- 100% of planned automated test cases executed, except those that cannot be executed in a course project (`TC-OPS-01`, 99% monthly availability), which are closed by agreement with the course instructor.
- 100% of functional requirements (`SYS-F-001..020`) verified and passed.
- 0 critical or high-severity open defects.
- Performance tests confirm 95% API response time < 800 ms and 200 concurrent transaction capability.
- Security verification confirms TLS 1.3, bcrypt hashing, 48h JWT expiration, and zero SQLi/XSS vulnerabilities.
- Automated code coverage meets or exceeds 70%.

5.1 Security Validation

- **Credential & Password Security:** Validate temporary credential generation, forced password reset on first login, and bcrypt password hashing (factor 12) (`PRJ-SR-002`).
- **Session Security:** Verify HTTP-only JWT token issuance, payload signing, and automated token invalidation post-48 hours (`PRJ-SR-003`).
- **Transit Encryption:** Ensure TLS 1.3 encryption is strictly enforced across all REST API endpoints (`PRJ-SR-001`).
- **Access Control (RBAC):** Execute role matrix verification ensuring Student, Faculty, and Admin role permissions are strictly checked by middleware (`PRJ-SR-004`).
- **Input Fuzzing & SQLi Protection:** Validate parameterized ORM queries against SQL injection and cross-site scripting (XSS) using OWASP ZAP automated scanners (`PRJ-SR-005`).
- **Automated Freeze Lock:** Verify backend API time validation rejects Faculty attendance edits exactly 48 hours post-session start (`SYS-F-010`).

6. Test Environment

Hardware

- **Application & DB Server:** Docker Compose stack (MySQL 8, the Flask API on gunicorn, and the nginx web tier) running on developer laptops with Docker Desktop and on GitHub-hosted Linux runners in CI.
- **Client Devices:** Workstations and mobile screens (viewport widths 375px to 1920px).

Software

- **Application Stack:** Python 3.12 (Flask, SQLAlchemy) served by gunicorn, React frontend served by nginx, MySQL 8.
- **OS Environment:** Linux containers (`python:3.12-slim` for the API, `nginx:1.27-alpine` for the web tier, `mysql:8.0` for the database); Ubuntu on the CI runners.

Tools

- **Unit & Integration Testing:** PyTest against a real MySQL 8 database (backend) and Vitest (frontend logic).
- **API Testing:** Newman CLI test collection runner.
- **Performance Testing:** Apache k6 load testing framework.
- **Security Scanning:** OWASP ZAP vulnerability scanner (API scan, API baseline and UI baseline), Bandit (static analysis of the Python source), pip-audit and npm audit (known-vulnerable dependencies).
- **UI & E2E Testing:** Playwright automation framework, with axe-core for WCAG 2.1 A/AA accessibility checks.
- **Automation:** every tool runs in Docker through the `qa.sh` stages, and GitHub Actions runs them on every pull request and on every push to `main`.

Test Data

- Synthetic database seed containing 500 student accounts, 50 faculty accounts, 10 admin accounts, 50 course offerings, and 1,000 scheduled session records. The database is re-seeded

before each black-box stage, so every stage starts from identical data.

7. Test Schedule

Testing activities are integrated directly into two 1-week Agile development sprints:

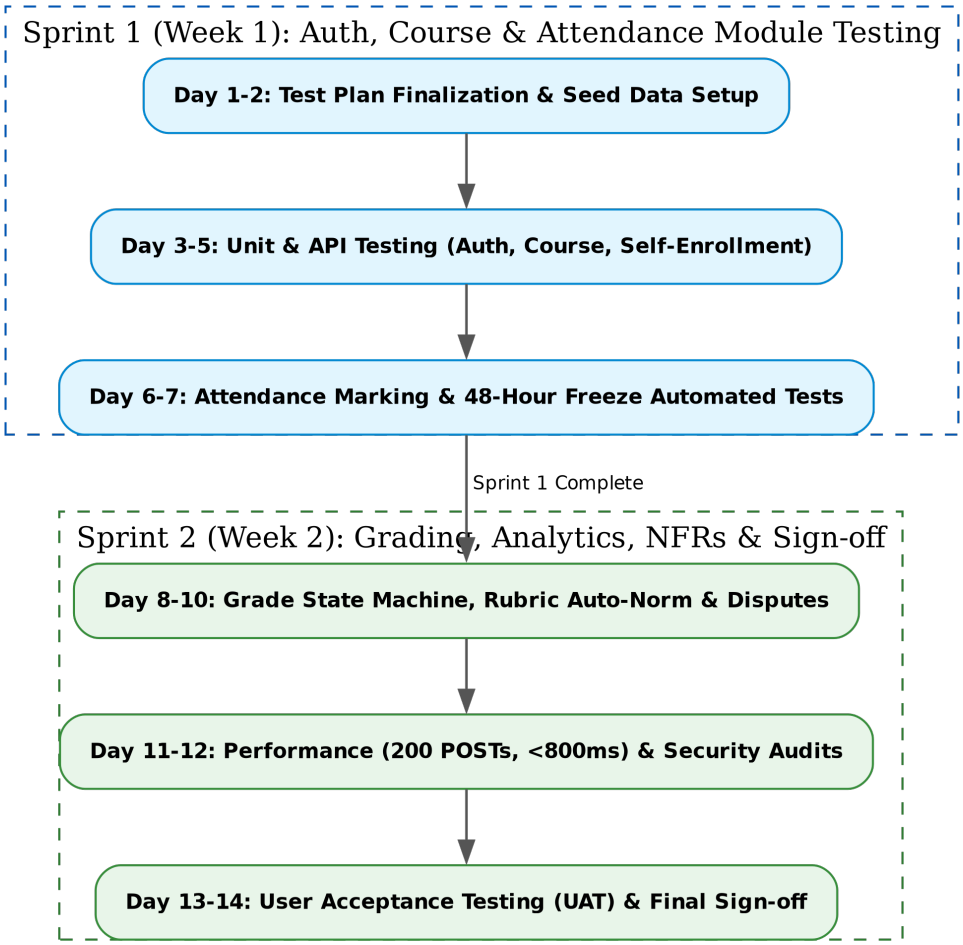


Figure 2: Agile Test Schedule Timeline

Milestones

- **Test Plan & Matrix Finalization:** 15-09-2026
- **Test Environment & Seed Data Setup:** 18-09-2026
- **Sprint 1 Execution Start (Auth, Course, Attendance):** 20-09-2026
- **Sprint 1 Execution End:** 27-09-2026
- **Sprint 2 Execution Start (Grading, Analytics, NFRs):** 28-09-2026
- **Sprint 2 Execution End:** 05-10-2026
- **User Acceptance Testing (UAT):** 08-10-2026 to 10-10-2026
- **Final Sign-off:** 12-10-2026

8. Test Deliverables

- Software Test Plan (this document - `STP.pdf`).
- Automated Test Scripts (Unit, API, Performance, Security, E2E, UAT): `backend/tests`, the frontend unit tests, `qa/newman`, `qa/k6`, `qa/zap`, `qa/e2e` and `qa/uat`.
- Test Data Seed Scripts: `backend/app/seed`.
- Test Execution Logs and Coverage Reports: the raw tool output in `qa/reports/`, and the final unit, system validation, code coverage and branch coverage reports in `docs/reports/`.
- Defect Reports and Triage Summaries: `docs/qa/SECURITY-TRIAGE.md` and the defect log in `docs/uat/UAT-RESULTS.md`.
- Final Test Execution & Acceptance Report: `qa/reports/TEST-EXECUTION-REPORT.md`, with the acceptance results and sign-off in `docs/uat/`.

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Vibha Vasisht	Prepare plan, coordinate execution.
Test Engineer	Uttam B S	Design & execute test cases, log defects.
Developer	Tanmay T A	Support defect fixes and triage.
Course Instructor / Evaluator	Dr. Adithya Balasubramanyam	Approve test results, signoff readiness.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery in Sprint 1	Request early smoke builds and run automated unit tests on every pull request.

Risk	Mitigation
Test environment downtime or corruption	Maintain containerized Docker setup and automated database seed restore scripts.
Database deadlocks under 200 concurrent transactions	Execute early k6 load tests, optimize DB connection pooling and column indexing.

11. Assumptions & Dependencies

- MySQL database sandbox will be stable and available during execution.
- Synthetic test data (500 users, 50 courses) will be seeded prior to test execution.
- Development team will deliver feature modules according to the 2-sprint schedule.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment is unavailable for > 4 consecutive hours.
- Build is too unstable (blocking defects prevent execution of $> 30\%$ of planned test cases).
- Cascading database migration failures invalidate test data seeds.

Resume testing if:

- Blocking defects are resolved and verified with a smoke test.
- Environment is stabilized and synthetic test data is re-seeded.

13. Test Case Management & Traceability

Requirements Traceability Matrix (RTM) maps SRS requirement IDs to corresponding test cases and test suites:

SRS Requirement ID	Requirement Summary	Test Case ID	Test Suite
SYS-F-001	Account provisioning & credential display	`TC-ACC-01`	`TS-AUTH`
SYS-F-002	Mandate password reset on initial login	`TC-ACC-02`	`TS-AUTH`
SYS-F-003	Personal contact details update	`TC-ACC-03`	`TS-AUTH`
SYS-F-004	Admin account metadata/status update	`TC-ACC-04`	`TS-AUTH`
SYS-F-005	Course setup & rubric weight auto-norm	`TC-CRS-01`	`TS-CRS`

SRS Requirement ID	Requirement Summary	Test Case ID	Test Suite
SYS-F-006	Student self-enrollment (max 6 courses)	`TC-CRS-02`	`TS-CRS`
SYS-F-007	Class schedule creation & publishing	`TC-CRS-03`	`TS-CRS`
SYS-F-008	Restrict faculty roster visibility	`TC-CRS-04`	`TS-CRS`
SYS-F-009	Digitized binary roll-call submission	`TC-ATT-01`	`TS-ATT`
SYS-F-010	48-hour automated attendance freeze	`TC-ATT-02`	`TS-ATT`
SYS-F-011	Administrative attendance override	`TC-ATT-03`	`TS-ATT`
SYS-F-012	Student aggregate attendance view	`TC-ATT-04`	`TS-ATT`
SYS-F-013	Rubric score entry in `Draft` state	`TC-GRD-01`	`TS-GRD`
SYS-F-014	Submit draft grades with rubric weight	`TC-GRD-02`	`TS-GRD`
SYS-F-015	Admin grade publication	`TC-GRD-03`	`TS-GRD`
SYS-F-016	Student regrade request (`Under Review`, once per grade)	`TC-GRD-04`	`TS-GRD`
SYS-F-017	Resolve regrade dispute and re-publish	`TC-GRD-05`	`TS-GRD`
SYS-F-018	Student consolidated GPA dashboard	`TC-REP-01`	`TS-PERF`
SYS-F-019	Faculty class summary analytics	`TC-REP-02`	`TS-PERF`
SYS-F-020	Admin institution metrics export	`TC-REP-03`	`TS-PERF`
SYS-NF-001	Latency ≤ 800 ms at 95th percentile	`TC-PERF-01`	`TS-PERF`
SYS-NF-002	200 concurrent submission transactions	`TC-PERF-02`	`TS-PERF`

SRS Requirement ID	Requirement Summary	Test Case ID	Test Suite
SYS-NF-003	99.0% operational availability	`TC-OPS-01`	`TS-PERF`
SYS-NF-004	Responsive UI & ≤ 3 click roll-call	`TC-UX-01`	`TS-AUTH`
SYS-NF-005	3-tier separation & $\geq 70\%$ test coverage	`TC-QUAL-01`	`TS-AUTH`
PRJ-SR-001	Mandatory TLS 1.3 transit encryption	`TC-SEC-01`	`TS-SEC`
PRJ-SR-002	Bcrypt password hashing (factor 12)	`TC-SEC-02`	`TS-SEC`
PRJ-SR-003	48-hour HTTP-only JWT sessions	`TC-SEC-03`	`TS-SEC`
PRJ-SR-004	Endpoint RBAC middleware check	`TC-SEC-04`	`TS-SEC`
PRJ-SR-005	Parameterized ORM queries (SQLi/XSS)	`TC-SEC-05`	`TS-SEC`

14. Test Metrics & Reporting

Metrics collected

- % test cases executed: (Executed Test Cases/Total Planned) $\times$ 100%.
- % passed/failed: (Passed Test Cases/Executed Test Cases) $\times$ 100%.
- Defect density: Identified defects per 1,000 lines of code.
- Defect aging: Mean time from defect logging to resolution.
- Requirement coverage: (Verified Requirements/Total Requirements) $\times$ 100%.

Reports

- Daily execution status log.
- Final Test Summary & Acceptance Report.

15. Approvals

Role	Name	Signature / Date
QA Lead	Vibha Vasisht	Vibha Vasisht / 12-10-2026
Developer	Tanmay T A	Tanmay T A / 12-10-2026
Product Owner / Course Instructor	Dr. Adithya Balasubramanyam	Adithya B / 13-10-2026